

Problem Set 2

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Problem 1. Outlining Proofs(a),(b) State Machine: The set of state is $Q = \{\text{All possible permutations of } \{0, 1, 2, \dots, n-1\}\}$ **Start state:** Any one of the permutations of $\{0, 1, 2, \dots, n-1\}$.**Transition:**

$$(a_0, \dots, a_i, a_{i+1} \dots a_{j-1}, a_j, a_{j+1}, \dots, a_{n-1}) \rightarrow (a_0, \dots, a_j, a_{j-1} \dots a_{i+1}, a_i, a_{j+1}, \dots, a_{n-1})$$

IFF (i, j) is an inverted pair (i.e., $i < j$ but $a_i > a_j$).**Fianl state:** $q_f = (a_0, a_1, \dots, a_{n-1}) \mid (a_0 < a_1 < \dots < a_{n-1})$ which also implies that the list is sorted.(c) Derived variable: For any state $q = (a_0, a_1, \dots, a_{n-1})$, define

$$X(q) := a_0 \| a_1 \| \dots \| a_{n-2} \| a_{n-1}.$$

where $\|$ is the concatenation operator.

Also, $X(q)$ has only a nonnegative integer value because each term of $X(q)$ is part of state q . By definition each member of every valid state is defined over natural numbers, so the number obtained from concatenating natural numbers must also be natural.

We know that in any number, digits on the left are more significant than digits to the right of them, and during transition we swap terms a_i and a_j where $i < j$ and $a_i > a_j$, so after the swap, the value of the leftmost term taking part in the swap, which is at position i , will decrease, resulting in the overall value of the final concatenated value decreasing. This means $X(q)$ strictly decreases with each transition.

(d):

For the derived variable $X(q)$, from part (c), we know that its value is defined over natural numbers and $X(q)$ is strictly decreasing.

Then, by the termination theorem, we can conclude that the state machine will terminate, and from part(b) we know that the final state will be a fully sorted list.

Problem 2. A Fibonacci Sum

(a)

$$\begin{aligned}
 S &= \sum_{n=1}^{\infty} \frac{F_n}{c^n} \\
 S &= \frac{F_1}{c} + \frac{F_2}{c^2} + \frac{F_3}{c^3} + \frac{F_4}{c^4} \\
 \frac{S}{c} &= \frac{F_1}{c^2} + \frac{F_2}{c^3} + \frac{F_3}{c^4} \\
 \hline
 S - \frac{S}{c} &= \frac{F_1}{c} + \frac{F_2 - F_1}{c^2} + \frac{F_3 - F_2}{c^3} + \dots \\
 S - \frac{S}{c} &= \frac{F_1}{c} + \frac{F_0}{c^2} + \frac{F_1}{c^3} + \frac{F_2}{c^4} + \dots \quad [\because F(n) - F(n-1) = F(n-2)] \\
 S - \frac{S}{c} &= \frac{F_1}{c} + \frac{F_0}{c^2} + \frac{1}{c^2} \left[\frac{F_1}{c} + \frac{F_2}{c^2} + \frac{F_3}{c^3} + \dots \right] \\
 S - \frac{S}{c} &= \frac{1}{c} + \frac{S}{c^2} \quad [\because F_1 = 1, F_0 = 0]
 \end{aligned}$$

$$S = \frac{c}{c^2 - c - 1}$$

(b) By calculating the first 50 digits, I got,

0.00001000010000200003000050000800013000210003400055

Which magically has Fibonacci series represented by every 5 digits after decimal.

Problem 3. Integral Method

(a) Integral method requires function to be monotonically non-decreasing or non-increasing over the given range but $f(x)$ is violating that rule, by checking its first derivative $f'(x) = \frac{5-2x}{(5+x)^4}$, It indicates that function is non-decreasing until $x = 2.5$ than it becomes non-increasing.

(b)(c) Smallest i to approximate S_i can be found with first derivative of $f'(x)$ which is

$$f'(x) = \frac{5-2x}{(5+x)^4}$$

By putting $f'(x) = 0$, we get $x = 2.5$, and because $f'(2.5) > 0$, the function is increasing in the interval $[2.5, \infty)$. Meaning $i = 3$ is the minimum index that can be used to approximate the sum of S_i by the integral method.

$$S_0 = \sum_{x=0}^2 f(x) + \sum_{x=3}^{\infty} f(x)$$

$$S_0 = f(0) + f(1) + f(2) + S_3$$

Now we know from part (b) that S_3 is a non-increasing function, so by applying the integral method to find upper and lower bounds for S_3 .

$$\begin{aligned} S_3 &= \sum_{x=3}^{\infty} f(x) \\ I &= \int_3^n \frac{x}{(5+x)^3} \\ &= \lim_{n \rightarrow \infty} \left[\frac{5}{2(5+x)^2} - \frac{1}{(5+x)} \right]_3^n \\ &= \frac{11}{128} \end{aligned}$$

Now, for $f(x)$, which is non-increasing for $x > 2.5$ from the Integral bound theorem we know that

$$S_3 \in \left[\frac{11}{128}, \frac{47}{512} \right]$$

also we can calculate that $S_0 = f(0) + f(1) + f(2) + S_3 \in \left[\frac{114271}{1185408}, \frac{484867}{4741632} \right]$
accuracy achieved by my approximation is $\left[\frac{114271}{1185408} - \frac{484867}{4741632} \right] = \frac{3}{512}$.

(d) Take $j = 5$, and then by repeating the calculation from part (c) but taking $i = 5$ instead of 3, we get,

$$S_5 \in \left[\frac{3}{40}, \frac{2}{25} \right]$$

Now value of S_0 :

$$\begin{aligned} S_0 &= f(0) + f(1) + f(3) + f(4) + S_5 \\ &= 0 + \frac{1}{216} + \frac{2}{343} + \frac{3}{512} + \frac{4}{729} + S_5 \end{aligned}$$

Now to find u and v as rational numbers such that $u \leq S_0 \leq v$.

$$u = \frac{1}{216} + \frac{2}{343} + \frac{3}{512} + \frac{4}{729} + \frac{3}{40}$$

$$v = \frac{1}{216} + \frac{2}{343} + \frac{3}{512} + \frac{4}{729} + \frac{16}{200}$$

and in $S_0 = f(0) + f(1) + f(3) + f(4) + S_5$ we only approximated term S_5 , so error in S_5 is same as error in value of S_0 which is $\frac{2}{25} - \frac{3}{40} \leq \frac{1}{200}$ as required.

Problem 4. Some Sums

(a)

$$\begin{aligned} \sum_{i=0}^n \frac{5^i - 7^i}{13^i} &= \sum_{i=0}^n \left(\frac{5}{13}\right)^i - \sum_{i=0}^n \left(\frac{7}{13}\right)^i \\ &= \frac{1 - \left(\frac{5}{13}\right)^{n+1}}{1 - \frac{5}{13}} - \frac{1 - \left(\frac{7}{13}\right)^{n+1}}{1 - \frac{7}{13}} \quad \left(\text{using } \sum_{i=0}^n r^i = \frac{1 - r^{n+1}}{1 - r}\right) \\ &= \frac{13}{8} \left[1 - \left(\frac{5}{13}\right)^{n+1}\right] - \frac{13}{6} \left[1 - \left(\frac{7}{13}\right)^{n+1}\right] \\ &= -\frac{13}{24} - \frac{13}{8} \left(\frac{5}{13}\right)^{n+1} + \frac{13}{6} \left(\frac{7}{13}\right)^{n+1} \end{aligned}$$

(b)

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^k (3^j - i) &= \sum_{i=1}^n \sum_{j=1}^k 3^j - \sum_{i=1}^n \sum_{j=1}^k i \\ &= n \sum_{j=1}^k 3^j - k \sum_{i=1}^n i \quad (\text{evaluating the outer and inner constant sums}) \\ &= n \left(\frac{3(3^k - 1)}{3 - 1}\right) - k \left(\frac{n(n+1)}{2}\right) \quad (\text{applying GP and AP series formulas}) \\ &= \frac{3n(3^k - 1)}{2} - \frac{kn(n+1)}{2} \end{aligned}$$